

# Manantsoa Andriamazotomahefa

24 years old | manamanantsoa@gmail.com | +82 10 3283 1710 | Seoul, South Korea

## EDUCATION

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| <b>Chonnam National University</b><br>Master's in Chemistry, Specialization: Biochemistry<br>• GPA: 4.38/4.5 | Sep 2023 – Aug 2025<br>Gwangju, South Korea     |
| <b>Duksung Women's University</b><br>Korean Language Training                                                | Sep 2022 – Aug 2023<br>Seoul, South Korea       |
| <b>University of Antananarivo</b><br>Bachelor's in Chemistry<br>• GPA: 3.62/4                                | Mar 2018 – Oct 2021<br>Antananarivo, Madagascar |

## EXPERIENCE

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| <b>Graduate Research Student</b><br>Biochemistry Lab, Chonnam National University<br>• Investigated the DNA-binding properties of <i>E. coli</i> aspartate ammonia-lyase and its specific interactions with DNA using Fluorescence spectroscopy.<br>• Proposed a mechanism of the allosteric activation by metal ions of <i>E. coli</i> aspartate ammonia-lyase by identifying its allosteric site through AI-guided site-directed mutagenesis and kinetic assays. | Sep 2023 – Aug 2025<br>Gwangju, South Korea |
| <b>Intern</b> , Chemistry Lab, Institut Pasteur of Madagascar – Antananarivo, Madagascar<br>• Laboratory assistant in water quality control (qualitative analysis, risk assessment) and food analysis (finished product testing, regulatory compliance, toxic pollutant assessment).<br>• Characterization of pesticide extraction methods using GC/MS and LC/MS.                                                                                                  | Feb – May 2022                              |
| <b>Volunteer</b> , Y'Dago, Association of Young Volunteers – Antananarivo, Madagascar<br>• Collaborator on the project Y'Speak, serving as a volunteer English language teacher.<br>• Advocate for environmental protection.                                                                                                                                                                                                                                       | Oct 2021 – Jun 2022                         |

## CONFERENCES & PRESENTATIONS

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| <b>Korean Society for Molecular and Cellular Biology – Winter Conference 2025</b><br>Poster presentation: "Metal-Ion Dependency of Asparatase from <i>Escherichia coli</i> " | Jan 2025<br>Pyeongchang, South Korea |
| <b>FAOBMB 2025</b><br>Poster presentation: "Mutational Analysis Unveils the Metal-Ion Binding Site in <i>Escherichia coli</i> Aspartase"                                     | May 2025<br>Busan, South Korea       |

## SCHOLARSHIPS & AWARDS

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| <b>Global Korea Scholarship (GKS)</b><br>Awarded in 2022 by the Government of South Korea | Sep 2022 – Aug 2025 |
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## CERTIFICATIONS, SKILLS & INTERESTS

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| <b>Certifications:</b><br>• <b>IELTS</b> (Overall: 8.0, R: 8.5, L: 8.5, W: 8.0, S: 7.5) – 20/05/2024<br>• <b>DALF C1</b> – 12/02/2026<br>• <b>Test of Proficiency in Korean</b> level 6 – 18/12/2024 |
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**Key Skills:**

- **Technical Skills:** Cell culture, protein expression and purification, recombinant DNA technology, Polymerase Chain Reaction (PCR), Site-directed mutagenesis, enzyme assays, EMSA, GC/MS, HPLC, FPLC, NMR, UV-Vis and fluorescence spectroscopy, water quality analysis (DO, TDS, COD, BOD), volumetric titration, calorimetric analysis, and sample preparation and extraction techniques.
- **Bioinformatics Skills:** protein structural analysis (PyMOL, UCSF Chimera), molecular docking (AlphaFold, Chai-1), multiple sequence alignment (MAFFT, MUSCLE), phylogenetic analysis (IQ-TREE), MestReNova and TopSpin.
- **Computer Skills:** Microsoft Office, LaTeX, Origin (data analysis and figure preparation), Python, R, Adobe Illustrator, Linux (bioinformatics workflows).
- **Languages:** Malagasy (Native), English (Fluent), French (Fluent), Korean (Advanced).

**Interests:** programming, foreign languages, outdoor activities, and sports.